

Table 2.1 A Brief History of Market Microstructure Research and TCA	
Era	General Interest
1970–1990	Spreads, Quotes, Price Evolution, Risk Premium
1990–2000	Transaction Costs, Slippage, Cost Measurement, Friction
2000–2010	Algorithms, Pre-Trade, Black Box Models, Optimal Trading Strategies
2010–	Market Fragmentation, High Frequency, Multi-Assets, and Portfolio Construction

that is being incorporated into trading algorithms and black box models is based on many of the earlier groundbreaking works.

How exactly has market microstructure research evolved over time? First, research in the 1970s and 1980s focused mainly on spreads, quotes, and price evolution. In the following decade, 1990–2000, interest turned to transaction cost analysis but focused on slippage and cost measurement. The following period, 2000–2010, was marked with decimalization and Reg-NMS, and research turned to algorithmic trading. Studies focused on pre-trade analysis in order to improve trading decision and computer execution rules, and the development of optimal trading strategies for single stocks and portfolios. The 2010 era so far has been marked with market fragmentation, high frequency trading, algorithmic trading in multi-asset classes, and exchange traded fund research. And many portfolio managers are studying how best to incorporate transaction costs into stock selection and portfolio construction. These eras of market microstructure research are shown in [Table 2.1](#).

To help analysts navigate the vast amount of market microstructure research and find an appropriate starting point, we have highlighted some sources that we have found particularly useful. First, the gold standard is the research by Ananth Madhavan in two papers: “Market Microstructure: A Survey” (2000) and “Market Microstructure: A Practitioner’s Guide” (2002). *Algorithmic Trading Strategies* (Kissell, 2006) provides a literature review in Chapter 1, and *Algorithmic Trading & DMA* by Barry Johnson (2010) provides in-depth summaries of research techniques being incorporated into today’s trading algorithms for both order execution and black box high frequency trading. Finally, Institutional Investor’s *Journal of Trading* has become the standard for cutting edge academic and practitioner research.